PAPER ID-311109

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM III) THEORY EXAMINATION 2020-21

Time: 3 Hours **DIGITAL SYSTEM DESIGN** ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

Q no.	Question	Marks	CO
a.	Construct full adder using logic gates.	2	
b.	What is the concept of "setup" and "Hold" time?	2	
c.	What is difference between flip flop and latches?	2	
d.	What is difference between "Ripple Carry Adder" and Carry Look-ahead Geneartor?	2	
e.	How many flip flops are needed to implement a 32 bit register?	2	
f.	What is barrel shifter?	2	
g.	Which gates are called universal gates and why?	2	
h.	Differentiate between combinational logic circuit and sequential circuits?	2	
i.	Convert binary code(00011011) to gray code.	2	
j.	Implement a 4:1 multiplexer using 2:1 multiplexer.	2	

SECTION B

2. Attempt any *three* of the following:

Q no.	Question	Marks	CO
a.	Convert the following i. Hexadecimal equivalent of the decimal number 256 ii. Decimal equivalent of $(123)_9$ iii. 378.93_{10} to octal iv. Convert A3BH and 2F3H into binary.	10	
b.	Simplify using k-map to obtain a minimum POS expression: (A'+B'+C+D)(A+B'+C+D)(A+B+C'+D")(A'+B+C'+D')(A+B+C'+D)	10	
c.	State and Prove Demorgan's theorem.	10	
d.	For the clocked JK flip-flop write the state table, draw the state diagram and the state equation.	10	
e.	Design a BCD adder using two 4 bit addresses.	10	

SECTION C

3. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	With the help of a neat diagram, explain the working of a two-input TTL NAND gate	10	
b.	With the help of a neat diagram, explain the working of any two I. a CMOS inverter, II. a two input CMOS NAND gate III. a two input CMOS NOR gate	10	

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4. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	Define the following terms I. threshold voltage II. II. propagation delay III. III. power dissipation IV. IV. fan-in V. V. fan-out	10	
b.	Discuss Mealy and Moore FSM. What do you mean by excitation table?	10	

5. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	Explain the operation of FLASH ADC.	10	
b.	Explain the operation of successive approximation ADC. Discuss it merits and demerits.	10	

6. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	Design a sequential circuit with two flip flops, A & B and one input x. when x=0, the state of the circuit remains the same when x=1 the circuit passes through the state transitions from 00 to 01 to 11 to 10 back to 00 & repeat.	10	
b.	Explain 4bit Johnson counter with circuit diagram and waveforms.	10	

7. Attempt any *one* part of the following:

Q no.	Question	Marks	CO
a.	Design and implement a synchronous 3-bit up/down counter using JK flip-flops.	10	
b.	With a neat diagram explain the operation of R-2R DAC.	10	

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